

DEEP PATEL

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SUMMARY

AI Engineer with an M.S. in Computer Science and hands-on experience building LLM, RAG, AI agent, and transformer fine-tuning projects. Skilled in Python, FastAPI, vector search, reranking, citation-aware generation, and evaluation workflows for grounded AI systems.

EDUCATION

University of Oklahoma

Aug 2023 - May 2025

MS in Computer Science

Coursework: Natural Language Processing, Artificial Intelligence, Machine Learning, Transformer Models

Indian Institute of Information Technology (IIIT), India

Aug 2018 - May 2022

BTech in Computer Science and Engineering

RESEARCH PROJECTS

TRACE: Claim Verification & Cost Control Layer for LLM Systems [GitHub] June 2026 - Present

- Built a modular **LLM trust layer** with guarded input routing, risk classification, evidence retrieval, claim verification, and final answer repair.
- Implemented **CAGE**, a claim-verification workflow that extracts atomic claims, retrieves supporting evidence, checks citation support, and removes unsupported statements before final output.
- Designed **CostGuard** to control model routing, retrieval depth, verification depth, compression level, token usage, and per-query cost before generation and verification.
- Added **domain-scoped memory** with injection screening, verified-claim storage, TTL-based re-verification, and cache invalidation to keep stored knowledge safer and fresher.

EXPERIENCE

Machine Learning Engineer Volunteer

Aug 2025 - Present

Community Dreams Foundation, Remote, USA

- Supported machine learning application development by analyzing datasets, preparing data workflows, and identifying model improvement opportunities for community-focused projects.

Machine Learning Engineer

May 2022 - May 2023

Firenix Technologies, Pune, India

- Built **Python and PyTorch-based computer vision pipelines** for image classification, face recognition, and anomaly-detection workflows, reducing false alerts by 20
- Improved **data preprocessing and inference workflows** by standardizing image cleaning, resizing, batching, and prediction handling for more reliable model outputs.
- Supported **model testing and backend integration** by packaging prediction workflows into reusable scripts and API-ready components.

PROJECTS

Agentic Research Assistant

[GitHub] Jan 2026 - May 2026

- Architected a **planner-based research system** that routes user queries across document Q&A and web research workflows using FastAPI, tool execution, and shared trace logging.
- Developed a **document RAG and evaluation pipeline** with PDF ingestion, embeddings, reranking, citation-based generation, Tavily/arXiv/Wikipedia tools, token-cost tracking, retrieval metrics, and LLM-as-judge scoring.

Privacy-Aware Financial Agent Backend

[GitHub] Sep 2025 - Dec 2025

- Designed a **deterministic-first financial AI pipeline** that routes questions through intent detection, risk tiering, policy planning, compute, validation, grounding, and LLM synthesis.
- Scaffolded a **FastAPI and Pydantic backend** with modular sections for secure data access, specialist financial agents, PII handling, audit logging, memory, and observability.

TECHNICAL SKILLS

Languages: Python, SQL

AI / LLM Systems: Agentic AI, Agentic RAG, Multi-Agent Systems, Agent Skills, Tool Calling, Structured Outputs, MCP, Context & Memory Management, Human-in-the-Loop

Backend & Data: FastAPI, Pydantic, REST APIs, PostgreSQL/SQLite, Redis, Docker, Git

Evaluation & Observability: LLM Evaluation, LLM-as-Judge, Retrieval Metrics, Citation Coverage, Token/Cost Tracking, Latency Tracking, Trace Logging, Prompt Caching